Spatial representations emerge from optimizing temporal separation of activations in a hippocampus-inspired navigation agent without spatial supervision

When animals are put into a novel environment, they typically have no prior expectations about spatial contingencies such as reward locations or task requirements. Nevertheless, the hippocampus in mammals quickly builds a spatial representation within the first exposure. How can spatial representations emerge from sensory inputs and learning objectives that do not explicitly encode space? We investigate this question using a bottom-up computational architecture inspired by hippocampal physiology, exploring whether spatial organization can arise from the temporal structure of sensory experience and self-generated behavior, without position labels, path integration, maps, or spatial rewards.

We therefore simulate an artificial agent in a virtual environment receiving exclusively egocentric input. The actions of the agent are generated by a model that combines dentate gyrus (DG)-like sparse sensory events with CA3-like sequences implementing a memory buffer. CA3 records the temporal separation (TS) between DG events, that provides two complementary learning signals: a **"push"** signal that encourages DG events to have more TS, shaping their place fields to be distinct; and a **"pull"** signal that rewards faster transition policy between DG events, such that the TS becomes a better proxy of navigational distance. Employing the model, exploration of a continuous visual environment leads to the emergence of spatially informative DG activity without explicit information about its physical location or spatial distances. We explore the contribution of behavior and different pathways of the TS signal to the place code using an ablation approach and find that TS-weighted learning specifically for the push signal produces higher DG spatial information, less redundant response maps, and a broader spatial distribution of field peaks.

Together, our results demonstrate how spatial coding can emerge from the self-generated behavior, TS-weighted learning, and sequential memory, without explicitly optimizing for spatial information.

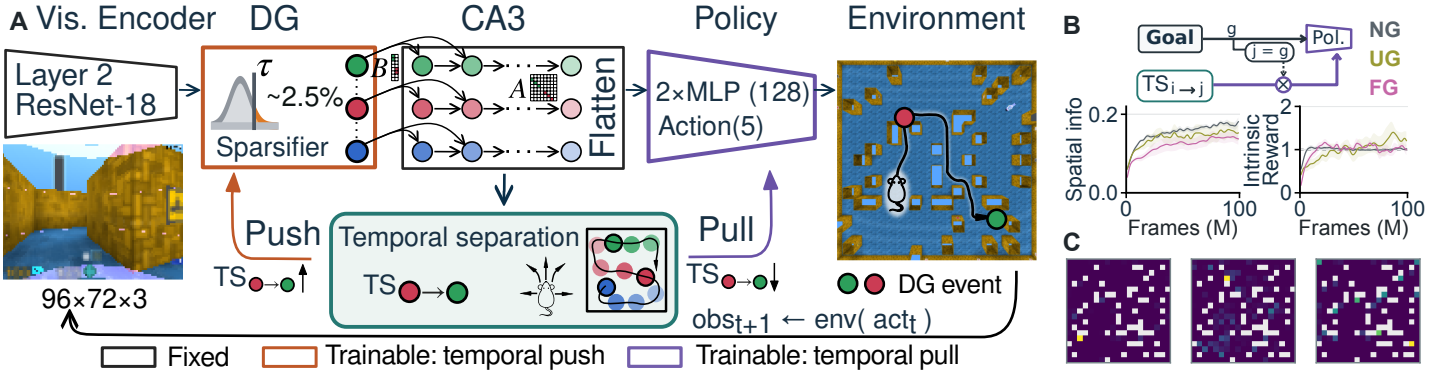


Figure 1: Temporal separation learning architecture. A, Egocentric visual input passes through a fixed ResNet-18 encoder, a sparse DG layer (approximately 2.5% activity), and fixed CA3-like sequence memory, followed by a policy readout selecting among five actions (arrows around the mouse). The illustrated matrices show CA3 recurrence (A) and DG input injection (B) for two DG units. Black outlines indicate fixed components; red and blue indicate DG (push) and policy (pull) learning pathways, respectively. The temporal separation (TS) between eligible DG events provides learning signals for both pathways. DG events on the map are conceptual: place fields are not prescribed, and the same unit may fire at different locations. **B,** Goal conditioning of policy and gating of intrinsic reward. NG: no goal conditioning, events from any DG units give intrinsic reward; UG: uniform goal selection, only rewards goal DG events; FG: frontier goal selection, favoring graph-frontier DG goals. A new goal is selected on goal hit or time out (CA3 memory length). Left: mean spatial information of DG units during training. Right: mean intrinsic reward per episode, normalized within each architecture. Lines show five-seed means with ± SEM. **C,** Example DG place fields from FG; gray bins were unvisited.

Learning objectives. We use temporal separation ($TS_{i\to j}$), measured in policy decisions between eligible DG events, as an intrinsic learning signal. The architecture and two complementary learning objectives are:

$$u_t = [\mathrm{BN}(Wf(o_t)) - \tau]_+, \quad x_{t+1} = Ax_t + Bu_t, \quad (\pi_t, V_t) = D_\theta(x_t, d_t) \tag{1}$$

$$\mathcal{L}_{\mathrm{DG}}(W) = -\mathrm{sg}(TS_{i\to j})u_{t,j}, \quad \mathcal{L}_{\mathrm{policy}}(\theta) = -\mathbb{E}_{\mathcal{T}\sim\pi_\theta}\left[\textstyle\sum_t \gamma^t (L - TS_{i\to j_t})\right] \tag{2}$$

In Eq. (1), $f(o_t) \in \mathbb{R}^{3840}$ denotes visual features, W the trainable DG projection, BN batch normalization, and $\tau = 2.43$ the sparsification threshold. The DG activity $u_t \in \mathbb{R}^{16}$ drives a CA3 shift register with $L = 71$ memory length per DG unit, giving a flattened CA3 state of $x_t \in \mathbb{R}^{L\cdot 16}$. Fixed matrices A and B implement sequential shifting and DG input injection (to the first 8 slots in each sequence independently; 2 visualized in 1), respectively.

The learned MLP readout D_θ receives CA3 state x_t and egocentric depth $d_t \in \mathbb{R}^{1\times 10}$ mimicking whisking sensation. In Eq. (2), the DG **"push"** reinforces eligible event activity $u_{t,j}$ proportionally to the preceding interval $TS_{i\to j}$, with sg denoting stop-gradient. Conversely, the policy **"pull"** rewards shorter event intervals relative to the CA3 memory length L, optimizing discounted intrinsic return, with advantage actor–critic and PPO modifications.

These opposing pressures may exploit the relationship between movement and event timing to organize DG activity spatially, without explicitly optimizing physical distances.

Spatial emergence and exploration. DG spatial information rises during training in all three agent designs, and individual FG response maps show localized activity (Fig. 1B,C). The goal-conditioned agents also take more direct paths than the no-goal agent in terminal online windows (Fig. 2A). Because each policy determines which parts of the environment are sampled, these online results reflect both the evolving network and its changing experience.

Behavior-dependent expression. We next freeze each trained network and compare the sensory experience generated by its learned policy with that generated by random actions. Learned actions yield higher DG spatial information in 36 of 38 models and more reliable DG maps in 37 of 38 (Fig. 2B). Thus, a fixed network expresses a stronger spatial signal under its own behavior. This comparison does not establish how the training policy shaped the network's weights.

Temporal DG push. To test whether event timing changes the learned representation itself, we compare temporal DG credit with mean-matched constant credit across five matched seeds per design. Replaying identical observations through each trained pair, temporal credit increases DG spatial information in all five no-goal pairs (mean +0.0575 bits per decision), four uniform-goal pairs, and three frontier-goal pairs (Fig. 2C). More consistently, it reduces overlap between DG response maps in all 15 pairs and broadens the spatial spread of eligible field peaks in 13. Temporal push therefore differentiates DG fields across space even when its effect on information within individual units varies. Reliable stored-graph edge counts, measured on separate policy-driven paths, do not change consistently.

Temporal policy pull. We separately compare temporal with constant target-hit credit while keeping temporal DG credit fixed in the two goal-conditioned designs (Fig. 2D). DG spatial information, map overlap, and peak spread show mixed effects across five matched seeds per design. Frontier-goal agents gain reliable stored-graph edges in all five pairs, whereas uniform-goal edge changes are mixed. These measurements use each policy's own trajectory, so they cannot separate changes in the network from changes in what it sees. The edges record event transitions in memory; whether the agent reaches commanded physical goals remains to be tested.

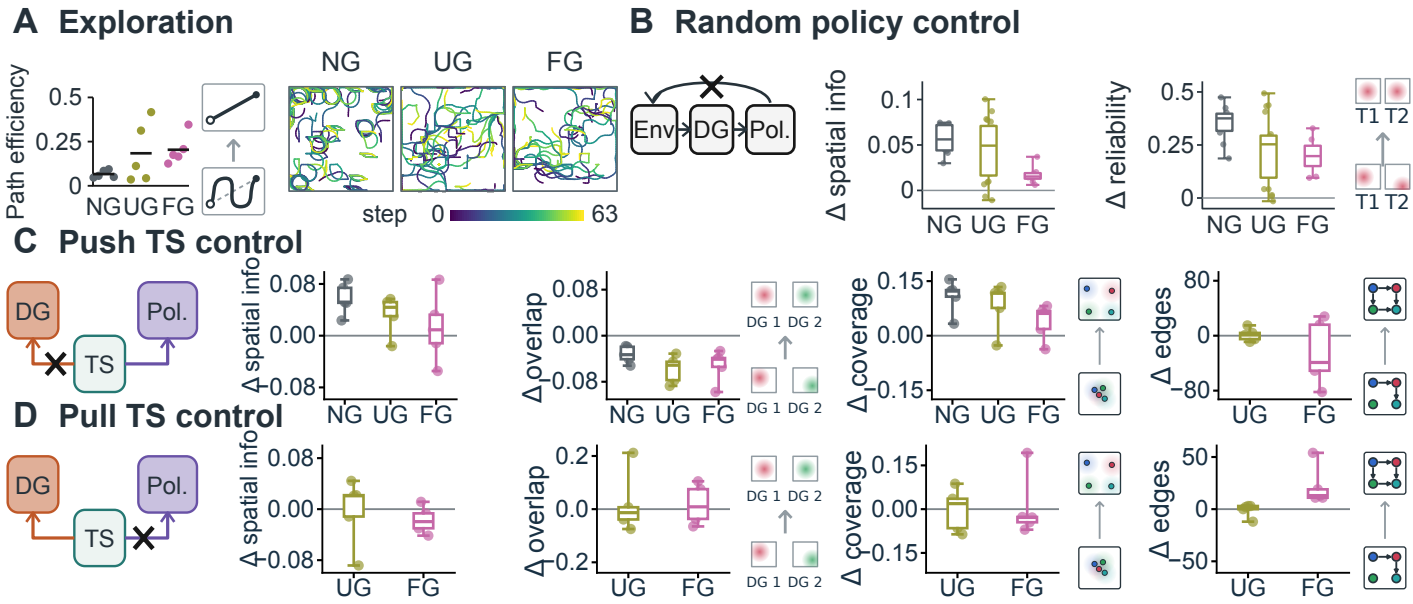


Figure 2: Exploration, behavior, and temporal-credit controls. **A,** Path efficiency for five terminal seeds per design, calculated as summed displacement across 64-decision trajectory fragments divided by distance traveled. Example tiles show 65 fragments per design; color indicates step within each fragment. **B,** Learned-action minus random-action differences in DG spatial information (bits per decision) and split-half DG-map reliability. Each of 38 terminal networks is frozen and evaluated with two paired 50,000-decision probes. **C,** Temporal minus mean-matched constant DG credit (push), five matched seeds per design. DG spatial information, active-map overlap, and field-peak coverage use the same 10,001 observations within each pair; reliable stored-graph edges use separate policy-driven paths. **D,** Temporal minus constant target-hit credit (pull), five matched UG and FG seeds with temporal DG credit fixed. Measures use separate 100,000-decision policy-driven paths at $\gamma = 0.999$ and GAE $\lambda = 0.995$. Overlap is mean cosine between active DG maps; peak coverage describes how widely eligible field peaks span visited space. Reliable edges are stored directed event transitions, not verified goal arrivals. For B–D, dots show paired differences, boxes show quartiles and medians, whiskers show observed extrema, and gray lines mark zero. Icons illustrate higher raw metric values, not the direction of an observed effect.